

7.4 Integration of Rational Functions by Partial Fractions - Continued

RECALL: The Method of Partial Fractions

Consider the rational function, $f(x) = \frac{h(x)}{g(x)}$.

1. The degree of the numerator must be _____ the degree of the denominator. If this condition is not met, we must use _____ to accomplish it.

2. Factor _____, $g(x)$.

3. Let $x - r$ be a linear factor of $g(x)$. Suppose $(x - r)^m$ is the highest power of $x - r$ that divides $g(x)$. To this factor, assign the sum of m partial fractions

$$\frac{A_1}{(x - r)} + \frac{A_2}{(x - r)^2} + \frac{A_3}{(x - r)^3} + \dots + \frac{A_m}{(x - r)^m}$$

4. Let $x^2 + px + q$ be an irreducible quadratic factor of $g(x)$ so that $x^2 + px + q$ has no real roots (where a check for this is that _____). Suppose that $(x^2 + px + q)^n$ is the highest power of this factor that divides $g(x)$. Then to this factor, assign the sum of the n partial fractions

$$\frac{B_1x + C_1}{(x^2 + px + q)} + \frac{B_2x + C_2}{(x^2 + px + q)^2} + \dots + \frac{B_nx + C_n}{(x^2 + px + q)^n}.$$

5. Set the original fraction equal to the sum of all partial fractions.
6. Solve for _____.
7. Integrate.

Example 1

Evaluate $\int \frac{y^2 + 4y + 4}{y(y^2 + 2y + 2)} dy$

Example 2

Evaluate $\int \frac{\sec^2(x)}{\tan(x)(\tan(x) + 1)} dx$

Extra Examples

1. Evaluate $\int \frac{x+2}{x(x+1)^3} dx$

2. Evaluate $\int \frac{1}{1 + e^x} dx$